Yasaman BASAKHA

yasaman_basakha@yahoo.com • +33 6 87 16 34 96 • yasaman-basakha • Grenoble, France

Motivated materials scientist with dual Master's degrees in Polymer Engineering and Sustainable Biobased Materials. Skilled in composite formulation, polymer processing (injection molding, extrusion), and material characterisation (SEM, tensile testing). Currently researching cellulose-based composites for sustainable packaging at Grenoble-INP, Pagora. Passionate about advancing durable, high-performance materials.

EDUCATION

Grenoble INP - Pagora, UGA, Grenoble, France

Master • ***Materials Science and Engineering Program Biorefinery and Biomaterials*** • *Sep 2024 - Present*

Research Oriented - Year 2

Scholarship: Recipient of the International Excellence Scholarship sponsored by the company Albea

Supervisors: Julien BRAS, Emilien FRÉVILLE

Amirkabir University of Technology (Tehran Polytechnic), Tehran, Iran

Master • ***Polymer Engineering*** • *Sep 2016 - Feb 2019*

GPA: 3.5/4.0 - Thesis Grade: 4.0/4.0

Thesis: Evaluation of the effect of polyester/PEG-based hydrogels structure on their in-vitro biodegradation

Supervisors: Gity Mir Mohamad Sadeghi

Islamic Azad University, Science and Research Branch, Tehran, Iran

Bachelor • ***Polymer Engineering*** • *Sep 2011 - Jun 2015*

GPA: 3.45/4.0 - Thesis Grade: 4.0/4.0

Thesis: Evaluation of the suitable and biodegradable alternative for 3D printer filaments

Supervisor: Milad Mehranpour

EXPERIENCE

Laboratoire LGP2, Pagora, ***Research Intern***

Feb 2025 – Present

Grenoble, France

Internship Title: Injection-molding of cellulose-based material for packaging applications to replace single-use plastics.

- Formulated and processed sustainable composites (benchmark, DOE) to produce prototypes by injection-molding made of 90-95 % of cellulose content.
- Development of patented processes at pilot scale and industrial trials.
- Conception of molds for new packaging application et mechanical tests.
- Material characterization (mechanical and recycleability).
- End of life (recyclability, compostability, life cycle assessment).

Nanoalvand Pharmaceutical Company(CinnaGen Holding), ***Process Validation Expert***

2020-2022

Tehran, Iran

- Ensure compliance with regulatory requirements and industry standards
- Analyze data and prepare reports on validation activities
- Work cross-functionally to identify and resolve process issues
- Participate in audits and stay up-to-date with industry best practices and regulatory changes
- Temperature Mapping and preparing reports and documentation
- Prepare Process Validation reports
- Familiar with International GMP guidelines
- Holding weekly meetings for Cleaning Validation
- Analyze data and prepare reports on validation activities

Amirkabir University of Technology, ***Research Assistant***

2016-2019

Tehran, Iran

- Synthesis of a triblock copolymer utilizing PEG, specifically PCL-PEG-PCL
- Proficiently conducted the synthesis of Polyurethane Hydrogel
- Possess expertise in analyzing various techniques including FTIR, ATR, NMR, DMTA, and DSC
- Demonstrated capability in the production of Artificial Leather
- Responsible for designing experiments, overseeing product testing and evaluation, and managing laboratory operations.

Masterbatch and Compound Association , ***Secretary***

2019 - 2020

Tehran, Iran

- Managed administrative and secretarial duties, coordinated events, and maintained records
- Assisted in policy development, prepared correspondence and reports for the Board of Directors
- Responded to member inquiries with confidentiality and discretion.

Online and In-Person Private Tutor

2011 -2023

- Teach math and physics to high school students online and in-person
- Create lesson plans and learning materials, provide individualized feedback and support

TECHNICAL SKILLS

Polymer Synthesis
- Synthesis of triblock copolymer (PCL-PEG-PCL)
- Polyurethane hydrogel preparation
- Artificial leather production

Polymer & composite Processing
- Pulp formulation and enzymatic treatment
- Twin-screw extrusion
- Thermopress operation
- Injection molding (test scale)

Characterization & Testing
- Mechanical Testing (INSTRON- tensile, compression, bending tets)
- Microscopy and Morphological analysis (SEM, Optical microscopy, MorFi analysis)
- Chemical characterization technique (FTIR, ATR, NMR)
- Thermal analysis (DMTA, DSC)
- Recyclability evaluation (Aticelca method)

Material Design & Formulation
- Composite formulation with natural fillers (e.g., walnut shell, $CaCO_3$)
- Experience with both thermoplastic and biopolymer matrices

Laboratory & Experimental Design
- Experiment planning and execution
- Product performance evaluation
- Lab operations and material testing supervision

SELECTED COURSES

Master • ***Materials Science and Engineering Program Biorefinery and Biomaterials*** • *Sep 2024 - Present*
- Second Generation Biopolymers- 19/20
- Chemicals Recovery in Biorefinery mills- 18/20
- Polysaccharides Science and Characterization- 17/20
- Ecoconception, Lifecycle Analysis- 17/20
- Biogas Production - Methananisation- 17/20
- Biobased Materials - Project & Labworks- 16/20
- Biorefinery- 15.8/20

Master • ***Polymer Engineering***
- Research design- 19/20
- Polymer Degradation and Stabilization- 18.3/20
- Polymerization Reaction Engineering- 16/20
- Advanced Physical Chemistry of Polymer- 16/20

AWARDS AND SUCCESS

- *Top 75 out of ~4,500 applicants in the National Entrance Exam for Master's in Polymer Engineering, Iran (2016)*
- *Secured a position among the top 10 students during Bachelor's program, (2015)*
- *Attained a position within the top 5% in the National University Entrance Exam for Bachelor's Degree, (2011)*
- *Completed studies at the National Organization for Development of Exceptional Talents (Sampad), (2003-2010)*

REFERENCES

•Julien BRAS
Professor Grenoble INP - Pagora, UGA
International Mobility Manager/Head of the Master Biorefinery and Biomaterials
Email: Julien.bras@grenoble-inp.fr

•Mohamed Naceur BELGACEM
Professor Grenoble INP - Pagora, UGA
Email: Mohamed-Naceur.Belgacem@grenoble-inp.fr

•Emilien FREVILLE
Postdoc student at Grenoble INP-Pagorast
Email: Emilien.Freville@grenoble-inp.fr

• Vahid Haddadi Asl
Department of Polymer and Color Engineering Amirkabir University of Technology (Tehran Polytechnic)
Email: haddadi@aut.ac.ir

• Saeed Pourmahdian
Department of Polymer and Color Engineering Amirkabir University of Technology (Tehran Polytechnic)
Email: pourmahd@aut.ac.ir

• Milad Mehranpour
Department of Polymer Engineering, Azad University, Science and Research Branch (ISRBIAU), Tehran, Iran.
Email: m.mehranpour@srbiau.ac.ir